How-To Library

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Join Two Tables with an Advanced Rule

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Abstract

In Persistent Data Masking and Data Subset, you can use an advanced rule to take input from two different tables and generate the output. This article describes how to create an advanced rule, join two tables with the rule, and edit the advanced rule.

Supported Versions

- Persistent Data Masking and Data Subset 9.5.2 HotFix 3

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Overview

You create an advanced rule to join two tables and generate a combined output. An advanced rule can receive input from two tables. You cannot use a standard or a custom rule to receive input from two tables.

You must add at least one input column and one output column when you create an advanced rule. You can also edit the advanced rule to change an input column or output column from required to optional.

To create an advanced rule and assign input from two tables to the rule, perform the following steps:

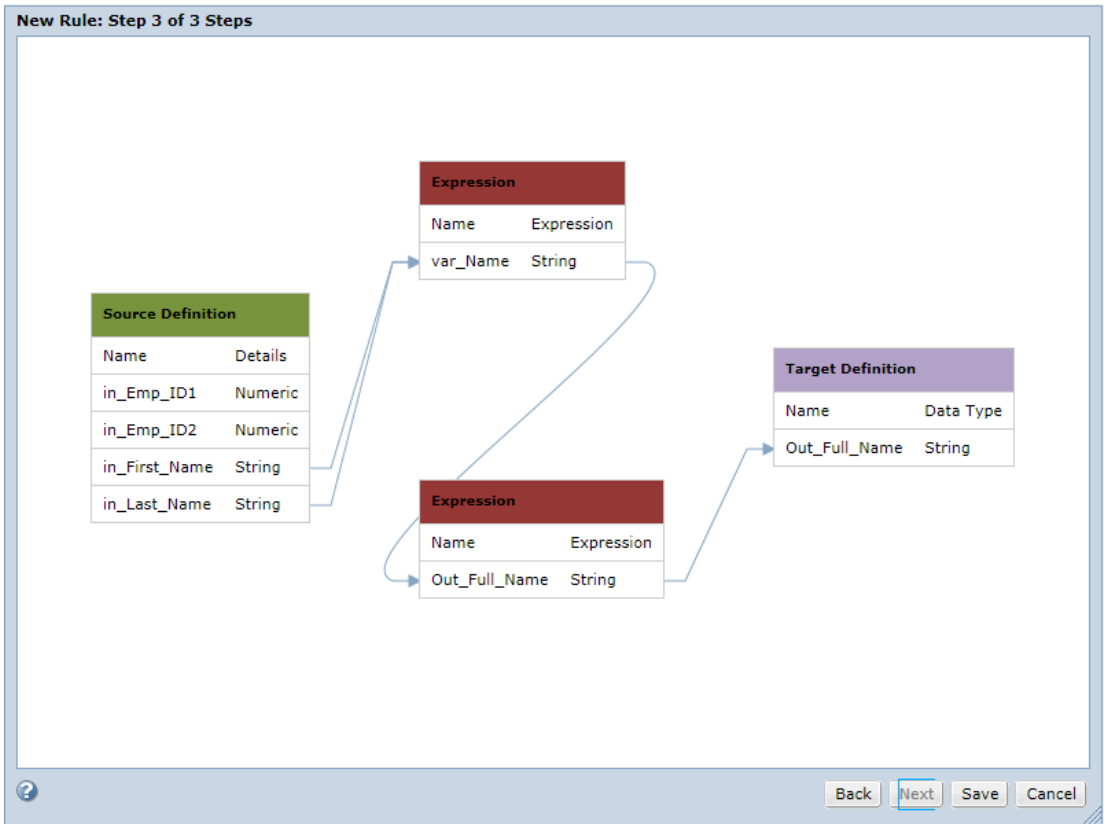
- Create an advanced rule with four input columns, one variable column, and one output column to accept input from two tables.
- Add the advanced rule to a project to assign masking rules to the columns.
- Add input from two tables to the advanced rule ports.
- Edit the advanced rule to mark an input column as optional.

Scenario

You work for an organization and you need to join first name and last name of the employees with an advanced rule. You create an advanced rule to combine first name and last name. If you apply standard masking on first name and last name, you can generate a masked output name.

Add input, output, and variable columns in the advanced rule. You select EMPLOYEES and EMPLOYEEDETAILS tables. Map the columns from the two tables to input and output columns of the advanced rule.

The following figure shows the relationship between the input, output, and variable columns in the rule map:



Step 1. Create an Advanced Rule

When you create an advanced rule, you create input, output, and variable columns. The input column contains two employee numbers, a first name, and a last name. The variable column contains a variable with expression to join the first name and the last name. The output column contains the combined name that is generated based on the variable column.

Creating Input Columns

Create an advanced rule and add input columns in the advanced rule.

1. In the **Policies** view, click **Actions** > **New** > **Rule**.

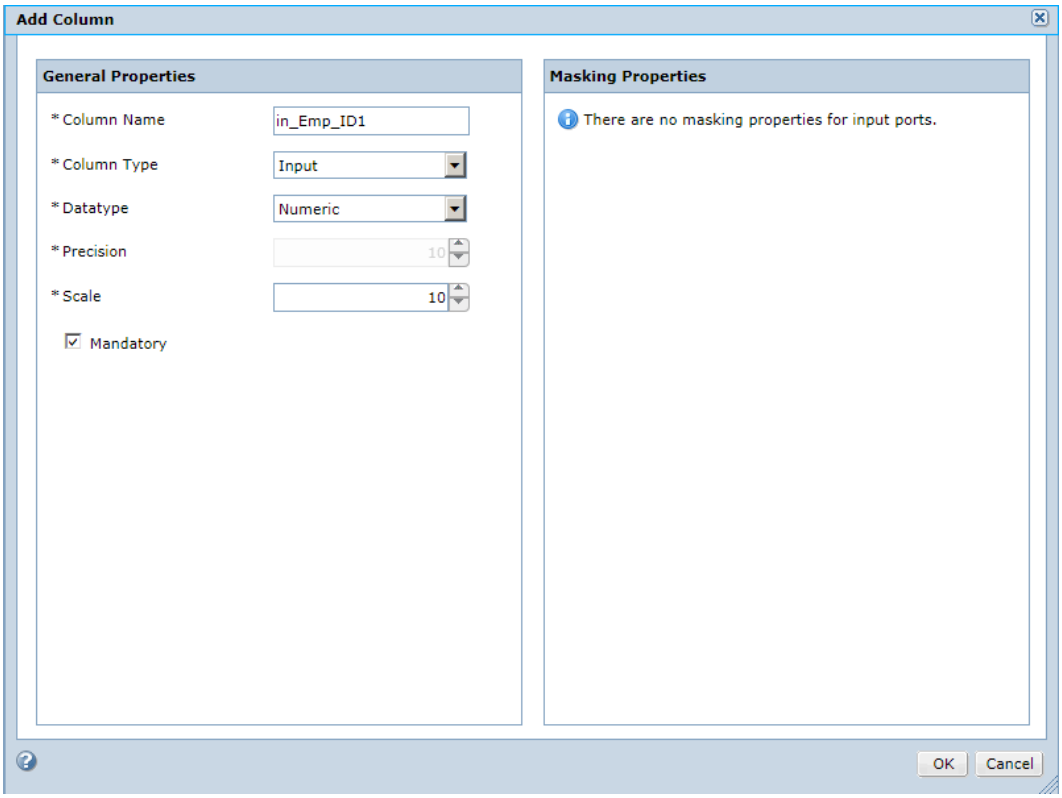
   The **New Rule** window appears.
2. Enter a name and an optional description for the rule.
3. Select the Masking Type as **Advanced**.
4. Click **Next**.
5. In the **Specify Masking Properties** window, click **Create Input Column**.

   The **Add Column** dialog box appears.

6. Under **General Properties**, configure the following properties:

Field	Value
Column Name	in_Emp_ID1
Column Type	Input
Datatype	Numeric
Precision	10
Scale	10
Mandatory	Enable the Mandatory option.

The following figure shows the in_Emp_ID1 column properties:



7. Click **OK**.

   The in_Emp_ID1 column appears in the list of Input Columns.

8. Click **Create Input Column**.

   The **Add Column** dialog box appears.

9. Under **General Properties**, configure the following properties:

Field	Value
Column Name	in_Emp_ID2
Column Type	Input
Datatype	Numeric
Precision	10
Scale	10
Mandatory	Enable the Mandatory option.

10. Click **OK**.

    The in_Emp_ID2 column appears in the list of Input Columns.

11. Click **Create Input Column**.

    The **Add Column** dialog box appears.

12. Under **General Properties**, configure the following properties:

Field	Value
Column Name	in_First_Name
Column Type	Input
Datatype	String
Precision	30
Scale	10
Mandatory	Enable the Mandatory option.

    You increase the precision value to avoid truncation of the string.

13. Click **OK**.

    The in_First_Name column appears in the list of Input Columns.

14. Click **Create Input Column**.

    The **Add Column** dialog box appears.

15. Under **General Properties**, configure the following properties.

Field	Value
Column Name	in_Last_Name
Column Type	Input
Datatype	String
Precision	30
Scale	10
Mandatory	Enable the Mandatory option.

You increase the precision value to avoid truncation of the string.

16. Click **OK**.

The in_Last_Name column appears in the list of Input Columns.

Creating Variable and Output Columns

Create variable column to join the first name and last name. Create an output column with first name and last name in the advanced rule.

1. Click **Create Input Column**.

The **Add Column** dialog box appears.

2. Under **General Properties**, configure the following properties:

Field	Value
Column Name	var_Name
Column Type	Variable
Datatype	String
Precision	30
Scale	10
Mandatory	Enable the Mandatory option.

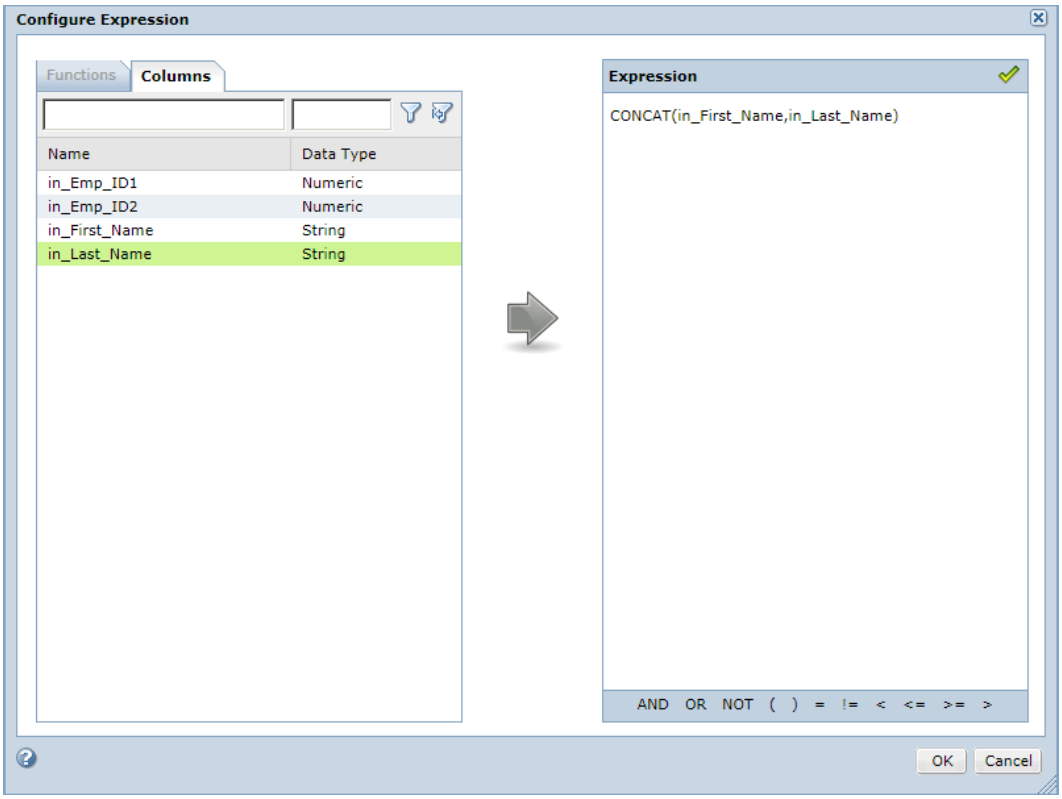
You increase the precision value to avoid truncation of the string.

3. Under **Masking Properties**, select **Expression**.
4. Click **Edit**.

The **Configure Expression** dialog box appears.

5. Click the **Columns** tab and enter the expression as CONCAT(in_First_Name,in_Last_Name).

The following figure shows the **Configure Expression** dialog box:

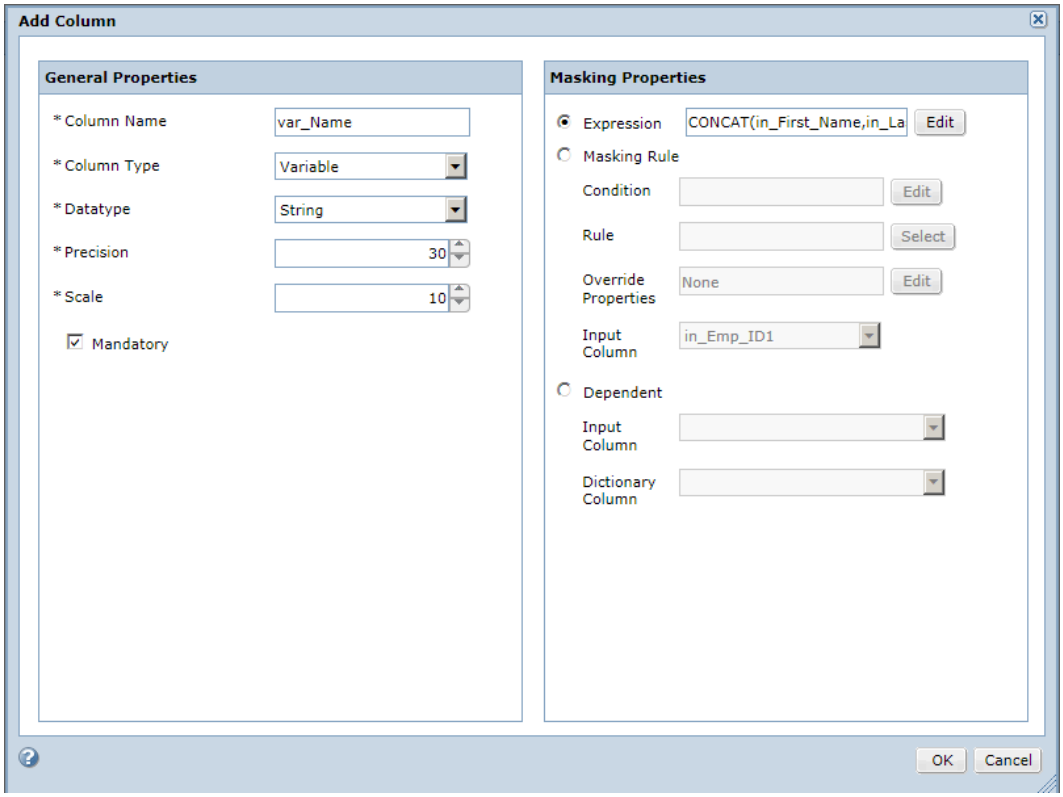


6. Click **Validate Expression**.

   An expression validated message appears. Click **OK**.

7. Click **OK**.

The **Add Column** dialog box with the var_Name column properties appears. The following figure shows the var_Name column properties:



8. Click **OK**.

   The var_Name column appears in the list of Variable Columns.

9. Click **Create Input Column**.

   The **Add Column** dialog box appears.

10. Under **General Properties**, configure the following properties:

Field	Value
Column Name	Out_Full_Name
Column Type	Output
Datatype	String
Precision	30
Scale	10
Mandatory	Enable the Mandatory option.

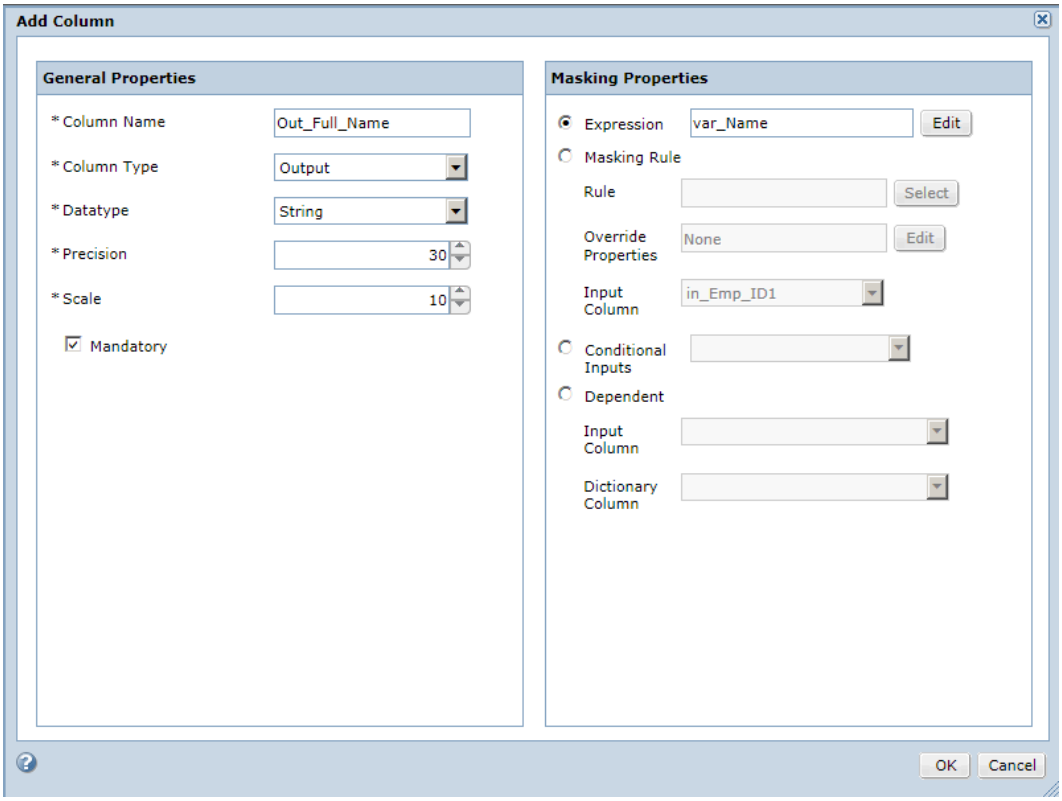
You change the precision value to avoid truncation of the string.

11. Under **Masking Properties**, select **Expression**.
12. Click **Edit**.

13. Enter the expression as var_Name.
14. Click **Validate Expression**.

    An expression validated message appears.
15. Click **OK**.

    The **Add Column** dialog box with the Out_Full_Name column properties appears. The following figure shows the Out_Full_Name column properties:



16. Click **OK**.

    The Out_Full_Name column appears in the list of Variable Columns.
17. Click **Next**.

    The rule map appears.
18. Click **Save**.

    The rule appears in the **Policies** view.

Step 2. Add the Advanced Rule to a Project

You open a project and add the advanced rule to the project. Once a rule is part of the project, you can use it to mask columns within the project.

1. Click **Projects**.

   You can see a list of projects.
2. Open a project from the list.

   The project window opens as another tab.
3. Click **Overview** > **Policies**.

4. Click **Actions** > **Add Additional Rules**.

   The **Add Additional Rules** dialog box appears.
5. Select the **Advanced_Rule1** from the list.
6. Click **OK**.

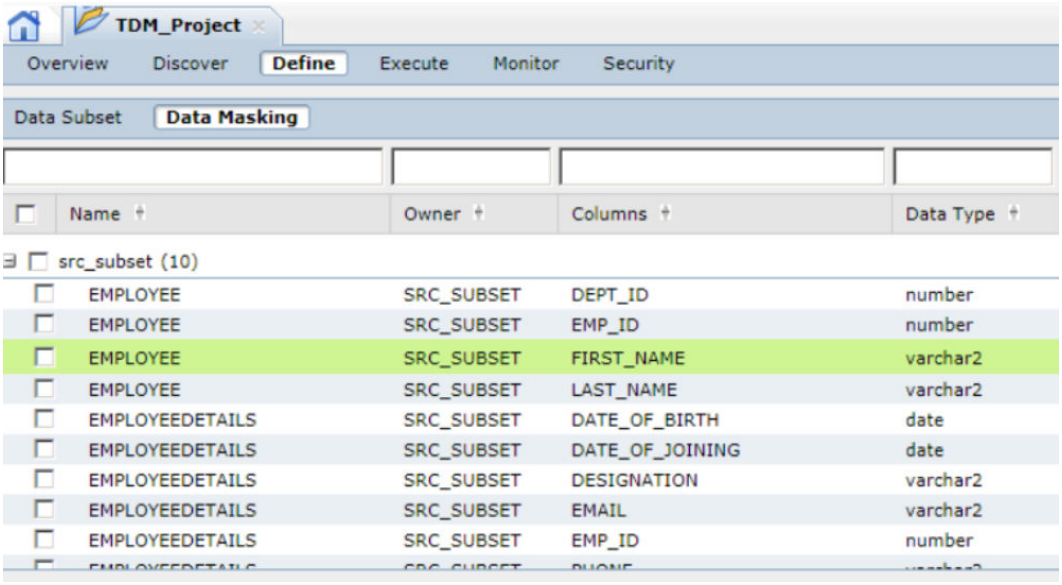
   The rule appears under the **Additional Rules** list.

Step 3. Join Two Tables with the Advanced Rule

Select two tables and assign the tables to the advanced rule. You can map a column from one table to the rule input port and the other table to the rule output port. This feature is not available in standard and custom masking rules.

1. Click **Define** > **Data Masking**.
2. Select First_NAME.
3. Under Rule column, select **Advanced_Rule1**.

   The following figure shows the list of tables:



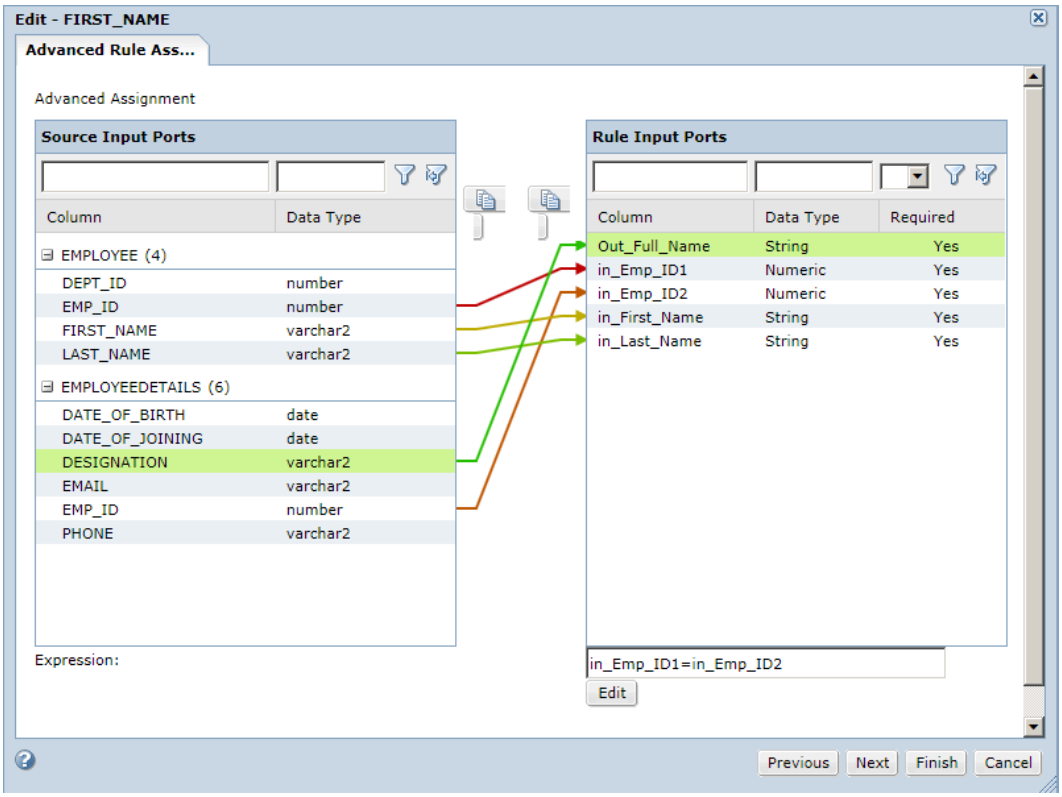
4. Click **Specify** under **Additional Details** column.

   The **Advanced Rule Assignment** window appears.
5. Select EMPLOYEE and EMPLOYEEDETAILS.
6. Click **Next**.

   The next screen displays the Source Input Ports and Rule Input Ports.
7. Map the columns from the EMPLOYEE and EMPLOYEEDETAILS tables to the input columns and the output columns under the Rule Input Ports.

   You must map all the required columns of the Rule Input Ports.
8. Click **Edit**.
9. Enter the join condition as in_Emp_ID1=in_Emp_ID2.
10. Click **OK**.

The following screen shows the mapping of the columns and the expression:



11. Click **Next**.

    The rule map appears.

12. Click **Finish**.

Step 4. Edit the Advanced Rule

When you create an advanced rule, all the columns appear as required by default. You can edit an advanced to rule. If you do not want to mask a column, you select an input or an output column to mark it as optional.

1. Click **Overview** > **Policies**.
2. Click **Advanced_Rule1**.

   The rule appears as another tab.

3. Click **Actions** > **Edit**.

   The **Edit-Advanced_Rule1** window appears.

4. Click **Masking Properties** .
5. In the **Input Columns**, select in_Emp_ID2, and click **Edit**.

   The **Edit Column** window appears with the Mandatory field selected.

6. Clear the **Mandatory** field.
7. Click **OK**.

   Under **General Properties**, the **Mandatory** field shows that it is not required.

8. Click **OK**.

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Acknowledgements

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